

Bike Data Analysis

1. Project Overview

This project analyzes **bike-sharing usage data** to understand rider patterns, revenue generation, and profitability. The analysis was conducted using **SQL Server Management Studio (SSMS)** by combining datasets from multiple years and calculating key business metrics.

The goal of this project is to:

- Combine yearly bike rental data
 - Calculate revenue and profit
 - Understand rider distribution across seasons, weekdays, and rider types
 - Generate insights that can support business decisions for bike-sharing services.
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2. Dataset Description

The project uses **three datasets**:

1 ☐ `bike_share_yr_0`

Contains bike rental data for the first year.

Main columns include:

- `dteday` – Date of rental
 - `season` – Season of the year
 - `yr` – Year indicator
 - `weekday` – Day of the week
 - `hr` – Hour of the day
 - `rider_type` – Type of rider
 - `riders` – Number of bike rentals
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2 ☐ `bike_share_yr_1`

Contains bike rental data for the second year with the same structure as year 0.

3 ☐ cost_table

Contains cost and pricing information.

Columns include:

- yr – Year
 - price – Rental price per ride
 - COGS – Cost of goods sold per ride
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3. Data Preparation

A **Common Table Expression (CTE)** was used to combine data from two years into a single dataset.

SQL Query Used

```
WITH cte AS (  
  SELECT * FROM bike_share_yr_0  
  UNION ALL  
  SELECT * FROM bike_share_yr_1  
)  
  
SELECT  
  dteday,  
  season,  
  a.yr,  
  weekday,  
  rider_type,  
  riders,  
  price,  
  COGS,  
  hr,  
  riders * price AS revenue,  
  riders * price - COGS AS profit  
FROM cte a  
LEFT JOIN cost_table b  
ON a.yr = b.yr
```

4. Data Transformation

Several transformations were applied during analysis:

1 ☐ Combining datasets

The `UNION ALL` operation merged rental records from both years into a single dataset for analysis.

2 **Joining cost data**

A **LEFT JOIN** was performed with the `cost_table` to attach pricing and cost information to each record.

3 **Calculated metrics**

Two key business metrics were created:

Revenue

`Revenue = Riders × Price`

Profit

`Profit = Revenue - Cost of Goods Sold`

These calculations help measure financial performance of the bike-sharing service.

5. Key Metrics Generated

The analysis produced the following important metrics:

- Total Riders
- Revenue generated per ride
- Profit per ride
- Seasonal bike usage
- Weekday vs weekend usage
- Hourly rental patterns

- Rider type distribution

SQLQuery1.sql - DE...\Anish (Work) (62) X

```
with cte as (  
    select * from bike_share_yr_0  
    union all  
    select * from bike_share_yr_1)  
  
select dteday,season,a.yr,weekday,rider_type,riders,price,COGS,hr,  
riders*price as revenue, riders*price-COGS as profit  
from cte a  
left join cost_table b  
on a.yr =b.yr
```

70 %

Results Messages

	dteday	season	yr	weekday	rider_type	riders	price	COGS	hr	revenue	profit
1	2021-01-01 00:00:00.0000000	1	0	6	casual	3	3.99	1.24	0	11.97	10.73
2	2021-01-01 00:00:00.0000000	1	0	6	casual	8	3.99	1.24	1	31.92	30.68
3	2021-01-01 00:00:00.0000000	1	0	6	casual	5	3.99	1.24	2	19.95	18.71
4	2021-01-01 00:00:00.0000000	1	0	6	casual	3	3.99	1.24	3	11.97	10.73
5	2021-01-01 00:00:00.0000000	1	0	6	casual	0	3.99	1.24	4	0	-1.24
6	2021-01-01 00:00:00.0000000	1	0	6	casual	0	3.99	1.24	5	0	-1.24
7	2021-01-01 00:00:00.0000000	1	0	6	casual	2	3.99	1.24	6	7.98	6.74
8	2021-01-01 00:00:00.0000000	1	0	6	casual	1	3.99	1.24	7	3.99	2.75
9	2021-01-01 00:00:00.0000000	1	0	6	casual	1	3.99	1.24	8	3.99	2.75
10	2021-01-01 00:00:00.0000000	1	0	6	casual	8	3.99	1.24	9	31.92	30.68
11	2021-01-01 00:00:00.0000000	1	0	6	casual	12	3.99	1.24	10	47.88	46.64
...	...	...	...	...	...	...	...	...	...	...	...

6. Business Insights

Based on the analysis, the following insights can be derived:

Seasonal Trends

Bike rentals tend to vary across seasons. Certain seasons show higher rider activity due to favorable weather conditions.

☐ Peak Usage Hours

Hourly analysis can reveal peak commuting hours when bike rentals are highest.

Rider Type Analysis

Different rider types contribute differently to total usage, helping businesses understand their primary customer base.

Revenue and Profit

By linking rider counts with pricing and cost data, the project estimates **revenue and profit per ride**, helping evaluate business profitability.

7. Tools Used

- **SQL Server Management Studio (SSMS)**
 - **SQL**
 - **Data Aggregation & Joins**
 - **Common Table Expressions (CTE)**
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8. Project Outcome

This project demonstrates how SQL can be used to:

- Combine datasets
- Perform data transformations
- Generate financial metrics
- Extract meaningful business insights from raw data.

The results of this analysis can help bike-sharing companies optimize operations, understand rider demand, and improve profitability.

Recommendation

Conservative Increase: Considering the substantial increase last year, a more conservative increase might be prudent to avoid hitting a price ceiling where demand starts to drop. An increase in the range of 10-15% could test the market's response without risking a significant loss of customers.

Price Setting:

If the price in 2022 was \$4.99, a 10% increase would make the new price about \$5.49.

A 15% increase would set the price at approximately \$5.74.

Market Analysis: Conduct further market research to understand customer satisfaction, potential competitive changes, and the overall economic environment. This can guide whether leaning towards the lower or higher end of the suggested increase.

Recommended Strategy:

Segmented Pricing Strategy: Consider different pricing for casual versus registered users, as they may have different price sensitivities.

Monitor and Adjust: Implement the new prices but be ready to adjust based on immediate customer feedback and sales data. Monitoring closely will allow you to fine-tune your pricing strategy without committing fully to a price that might turn out to be too high.

